Gagandeep Shivanna

gagan017ger@gmail.com linkedin.com/in/gagandeep-s017 github.com/GagandeepS017
+49 15560731320 Berlin, Germany portfolio.gagandeep.app

Profile

Backend engineer with **2+ years** of production experience at Vrize, building RESTful APIs and microservices in Java and Spring Boot, and an **M.Sc. in Computer Science** with a focus on Big Data and Artificial Intelligence, completed in 2026. Now builds LLM features in Python and FastAPI, from RAG pipelines and evaluation harnesses to a live text-to-SQL product, and wrote a master's thesis on how reliably LLMs write and repair infrastructure code.

Work Experience

Vrize India Pvt Ltd
09/2022 – 07/2024
Backend Engineer — *Bangalore, India*

- Built and maintained Java **Spring Boot** microservices and **RESTful APIs** for Zaxby's, a US restaurant chain, and Tonic, a healthcare client, over MySQL, PostgreSQL, and MongoDB.
- Traced failing endpoints with Postman, Swagger, and the debugger, fixed the exception handling behind them, and wrote **JUnit** and **Mockito** tests, resolving 50+ critical bugs.
- Automated build and deployment pipelines with **Docker**, Jenkins, and **Kubernetes**, reducing manual release work, and monitored the health of the services in Grafana.
- Gathered requirements directly from **client stakeholders**, presented delivered features to them in demos, and integrated third-party APIs with the in-house services.
- Tuned **SQL** queries and caching for payments, loyalty, and ordering features, and reviewed peers' pull requests across 15+ Agile sprints, recognized by senior engineers for technical ownership.

SRH University of Berlin
09/2025 – 04/2026
Research Assistant, NLP and Retrieval — *Berlin, Germany*

- Built end-to-end **RAG pipelines** with GPT and Claude for document classification and information extraction over research literature, served through **FastAPI** with Streamlit and React interfaces.
- Improved extraction accuracy over a zero-shot baseline by tuning chunking, embedding choice, and retrieval depth in **ChromaDB** against a held-out evaluation set.
- Worked across several projects under a professor, orchestrating multi-step workflows with **LangGraph** and enforcing structured outputs so extraction stayed machine-parseable across model versions.

Sovrano AI
05/2026 – 08/2026
AI Data Analyst Intern — *Remote, Germany*

- Hand-checked an automated LLM reviewer's scores on agent **tool-calling** tasks, adding human verification to model-graded evaluation.
- Wrote task decomposition specs and **annotation guidelines** for multi-step agent workflows, turning fuzzy requirements into precise steps and raising inter-annotator agreement.
- Authored gold-standard reference responses and ranked preference pairs for LLM fine-tuning and reward-model training, scored on accuracy, relevance, and safety.

Education and Training

SRH University of Berlin
10/2024 – 09/2026
M.Sc. Computer Science, Focus on Big Data & Artificial Intelligence (120 ECTS) – CGPA: 1.2 — *Berlin, Germany*

- **Master Thesis.** A failure taxonomy of **LLM-generated Infrastructure-as-Code**, running a four-referee verification pipeline (Terraform, Checkov, tfsec, OPA/Conftest) with an automated self-repair loop across 560 runs and three models. Survival analysis over 2,068 logged errors identifies which error categories resist repair, and a controlled ablation shows detailed error feedback causally drives repair success (McNemar $p < 0.0001$). Reproduced against the IaC-Eval benchmark, where repair lifts GPT-4o from 21% to 45%.
- **Research Paper (Ohrid Conference).** A stacking ensemble combining LSTM and Random Forest predictions in an XGBoost meta-learner, testing whether Tesla supplier stock movements carry predictive signal for Tesla's own price, achieving $\mathbf{R^2 = 0.9558}$ and a 14% RMSE reduction over the single-model baseline.

BMS College of Engineering, Bangalore
08/2018 – 07/2022

B.E. Information Science and Engineering – CGPA: 8.17 *Bangalore, India*

Skills

Programming	Python, Java, SQL, JavaScript/TypeScript
Development	Spring Boot, FastAPI, RESTful APIs, Microservices, PostgreSQL, MySQL, MongoDB, WebSockets
Cloud & DevOps	AWS, GCP, Docker, Kubernetes, Terraform, Jenkins, GitHub Actions, CI/CD
Testing	JUnit, Mockito, pytest, Checkov, tfsec
LLM & GenAI	RAG, Prompt Engineering, Claude API, OpenAI API, LangGraph, Ollama, Agent Workflows
Vector & Retrieval	ChromaDB, Embeddings, Semantic Search, Chunking Strategy
LLM Evaluation	Eval Harness Design, LLM-as-Judge Review, RLHF Preference Data, MLflow
AI/ML	PyTorch, Scikit-learn, Pandas, XGBoost, Transformers
Tools	Git, GitLab, Postman, Jira, Claude Code, Cursor

Projects

DataGrid, Production Text-to-SQL Analytics Platform
Live Demo | GitHub

FastAPI, React, DuckDB, Claude Sonnet/Haiku API, MLflow, SDV

- Deployed a **natural-language-to-SQL** platform where the model writes the query, DuckDB executes it sandboxed, and a second model streams the interpretation over SSE. Built a **text-to-SQL eval harness** measuring execution accuracy by result-set comparison rather than string match, logged to MLflow.

AsyncMeet (MANAN), AI Async Decision-Making Platform
Active Development

FastAPI, React/TypeScript, Claude Sonnet API, WebSockets

- A **SaaS platform** replacing live meetings, where a guided LLM chat generates its own follow-up questions to close gaps in participant input. Responses are synthesized into a structured decision summary on a real-time **WebSocket** dashboard.

AWS Account Onboarding Automation
GitHub

Terraform, GitHub Actions (OIDC), AWS, Python

- An **Infrastructure-as-Code** pipeline provisioning a full AWS account from a single YAML config, covering IAM, S3, and VPC, with **OIDC auth** and no static credentials. Its reference configs became the Tier B benchmark set in the thesis.

Certifications

- **Google Cloud**, MLOps for Generative AI, Core Infrastructure View certifications
- **AWS Educate**, Machine Learning Foundations, Cloud 101 View certifications

Language Skills

Languages: English (C1), German (B1)